COMPUTER VISION

Sport Video Analysis for Billiard Matches

Final Project Proposal

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1. Introduction

This proposal follows the final project brief for sport video analysis of Eight Ball billiard matches. The goal is to design a computer vision system that processes billiard video clips frame by frame and produces high-level information about the match state, including ball positions, ball categories, the playing-field area, and a 2D top-view minimap.

Our solution will be implemented in C++ with OpenCV and will use classical image-processing techniques rather than deep learning. The proposed system is intended to be robust across the provided benchmark clips, which differ in camera perspective, number of visible balls, and pool-table colour.

Project scope	**Classical computer vision system for analysing Eight Ball billiard videos frame by frame.**
Implementation constraint	C++ and OpenCV only; no deep learning models or external detector code.
Main outputs	Ball localisation/classification, playing-field boundary detection, semantic segmentation, and 2D top-view minimap.
Proposed extension	Pocket-aware game-state tracking that removes balls from the minimap when they are detected as pocketed.
Evaluation dataset	The 10 provided benchmark clips and the annotated first/last frames supplied with the project brief.

2. Required System Outputs

Task	Required output	Planned implementation
Ball recognition and localisation	Detect all balls and assign category IDs: cue ball, 8-ball, solid, or striped.	Hough circle detection inside the table mask, followed by HSV colour analysis.
Playing-field boundary detection	Detect the main lines delimiting the billiard playing field.	HSV cloth isolation, contour filtering, Canny edges, Hough lines, and line intersections.
Semantic segmentation	Generate a mask for background, cue ball, 8-ball, solid, striped, and playing field.	Layered OpenCV masks: background first, playing field second, ball classes last.
2D top-view minimap	Display ball positions and trajectories in a top-view representation for each frame.	Homography projection from image coordinates to a canonical table map with trajectory polylines.

3. Proposed Methodology

3.1 Playing Field Detection

For each clip, the table area will be detected from the first frame and reused for the remaining frames, following the project simplification that the camera pose does not change within a single video. The cloth region will be isolated through HSV thresholding and contour filtering. The upper rail edges will then be estimated using Canny edge detection and the Probabilistic Hough Transform. Intersections of the four selected boundary lines will define the table corners.

3.2 Homography and Minimap Coordinates

The four detected table corners will be mapped to a fixed top-view rectangle using cv::getPerspectiveTransform. This homography will convert ball centres and pocket centres from image space into minimap coordinates. It will also make trajectory drawing independent of the camera perspective.

3.3 Ball Detection and Classification

Ball candidates will be detected with cv::HoughCircles restricted to the playing-field mask. The classification step will examine HSV pixels inside each circle: low saturation and high value for the cue ball, low value for the black 8-ball, a dominant saturated hue for solid balls, and a combination of near-white and saturated regions for striped balls.

3.4 Semantic Segmentation

The segmentation mask will be generated using deterministic drawing operations. Pixels outside the detected table will be labelled as background, pixels inside the table will be labelled as playing field, and each detected ball circle will overwrite the mask with its ball-category label. This directly matches the label IDs described in the project brief.

3.5 Tracking and Pocket-Aware Extension

Consecutive detections will be associated using nearest-neighbour matching. Recent positions will be stored to draw trajectories for moving balls. As an additional contribution, the six pocket regions will be inferred from the table geometry. If a ball disappears near a pocket region, it will be marked as pocketed and removed from the active minimap state. If it disappears away from a pocket, it will be treated as an occlusion and kept temporarily at its last known position.

4. Assumptions and Robustness Plan

- The camera pose is assumed to remain fixed during each video clip, as allowed in the brief.
- The table boundary will use the visible upper rail/cloth attachment lines, consistent with the example provided by the professor.
- Different table colours will be handled by using configurable HSV ranges and validating the largest plausible table contour.
- Players, cue sticks, and spectators will be rejected where possible through table masking, geometric constraints, and temporal consistency.
- Additional internet or self-recorded videos will only be used for development if their source and annotation format are clearly documented in the final report.

5. Performance Measurement

The evaluation will follow the metrics specified in the project brief. The provided ground truth will be used only for evaluation, not as input to the algorithm.

Output	Metric	How it will be reported
Ball localisation	mAP at IoU threshold 0.5	Per-video results for frame_first.png and frame_last.png, with bounding-box visualisations.
Segmentation	mIoU averaged over background, cue ball, 8-ball, solid, striped, and playing field	Per-class IoU and average mIoU, with predicted masks shown beside the input frames.
2D minimap and trajectories	Qualitative comparison with the visible final game state	Final minimap for each clip, including ball positions, movement trajectories, and pocketed-ball state where applicable.

6. Work Division and Estimated Hours

Member	Main contribution	Estimated hours
Vishesh Goyal	Playing-field detection, boundary-line selection, homography, pocket geometry, and minimap rendering.	35 h
Himashi Bogahawaththa	Ball detection, HSV classification rules, semantic segmentation, tracking logic, and pocket-aware state update.	35 h
Both members	Integration, CMake configuration, benchmark runs, result tables, report writing, and final video generation.	15 h each

The total estimated effort is approximately 100 person-hours. The final report will clearly identify each member's contribution in ideas, implementation, testing, and performance measurement, as requested in the project brief.

7. Development Milestones

Milestone	Focus	Expected output
1	Table boundary and homography	Stable playing-field corners and top-view mapping for each clip.
2	Ball detection, classification, and masks	Bounding boxes, class labels, and segmentation masks for benchmark frames.
3	Tracking and minimap overlay	Ball trajectories, pocket-aware active state, and output videos with minimap.
4	Evaluation and reporting	mAP, mIoU, qualitative figures, error analysis, and final report content.

8. Planned Delivery

- C++ source code using OpenCV only, with one main author identified in each source file header.
- CMake configuration files and build instructions tested in the Virtual Lab environment.
- A final report with the proposed method, implementation details, quantitative metrics, qualitative output images, and error analysis.
- Output images for every benchmark frame pair, including bounding boxes and segmentation masks.
- Output videos for all benchmark clips with the 2D top-view minimap superimposed on the input video.
- A clear statement that the delivered code was written by the group members and does not include code generated by ChatGPT or similar tools.